Weekly project: Diabetes

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("daibb.csv")
print(df.head())

   Feature1  Feature2  Outcome
0  2.882026  1.663770        0
1  2.200079  1.820223        0
2  2.489369  1.593427        0
3  3.120447  1.136859        0
4  2.933779  2.088713        0

X = df.drop("Outcome", axis=1)
y = df["Outcome"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=10000)

model.fit(X_train, y_train)
```

LogisticRegression

▼ Parameters

penalty	'l2'
dual	False
tol	0.0001
C	1.0
fit_intercept	True
intercept_scaling	1
class_weight	None
random_state	None
solver	'lbfgs'
max_iter	10000
multi_class	'deprecated'
verbose	0
warm_start	False
n_jobs	None
l1_ratio	None

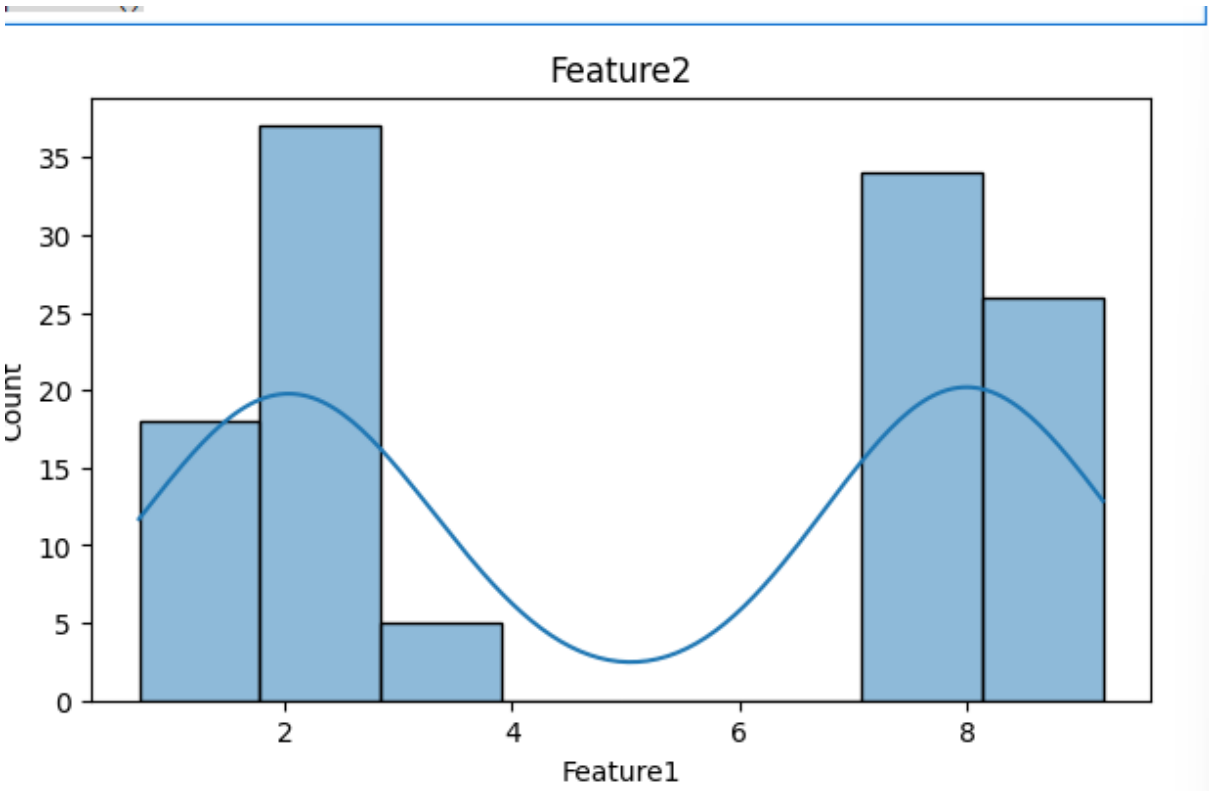
```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Logistic Regression Accuracy:", accuracy*100)
```

Logistic Regression Accuracy: 100.0

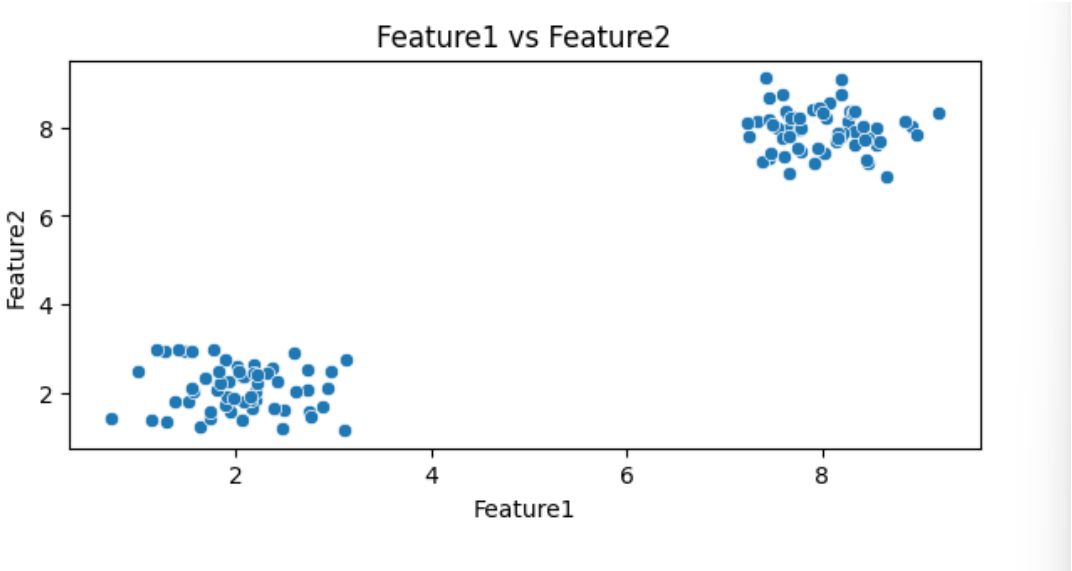
```
df = pd.read_csv("daibb.csv")
print(df.head())
```

	Feature1	Feature2	Outcome
0	2.882026	1.663770	0
1	2.200079	1.820223	0
2	2.489369	1.593427	0
3	3.120447	1.136859	0
4	2.933779	2.088713	0

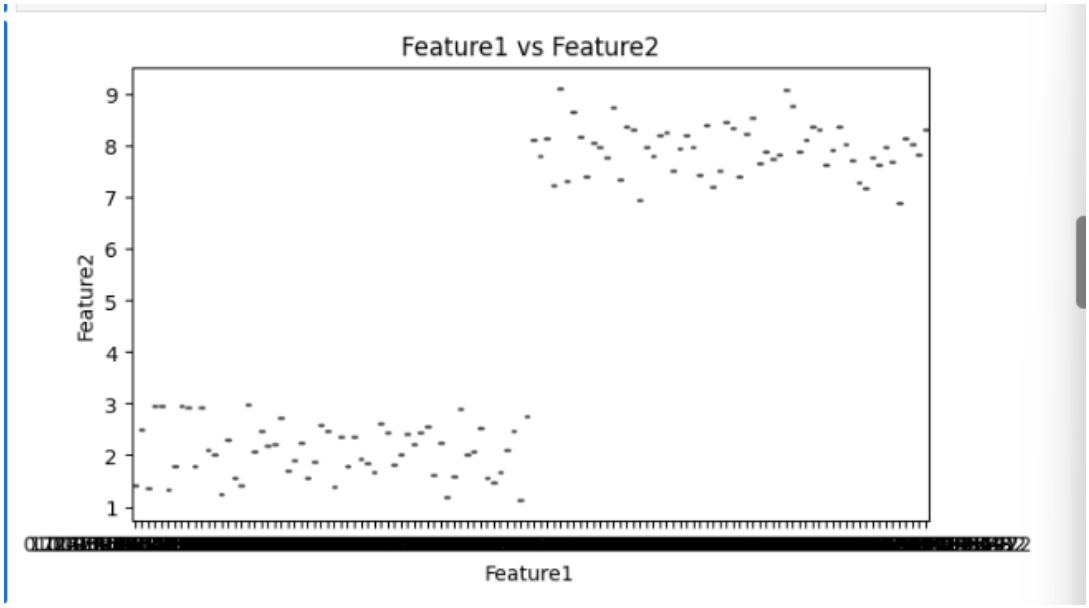
```
plt.figure(figsize=(7,4))
sns.histplot(df['Feature1'], kde=True)
plt.title("Feature2")
plt.show()
```



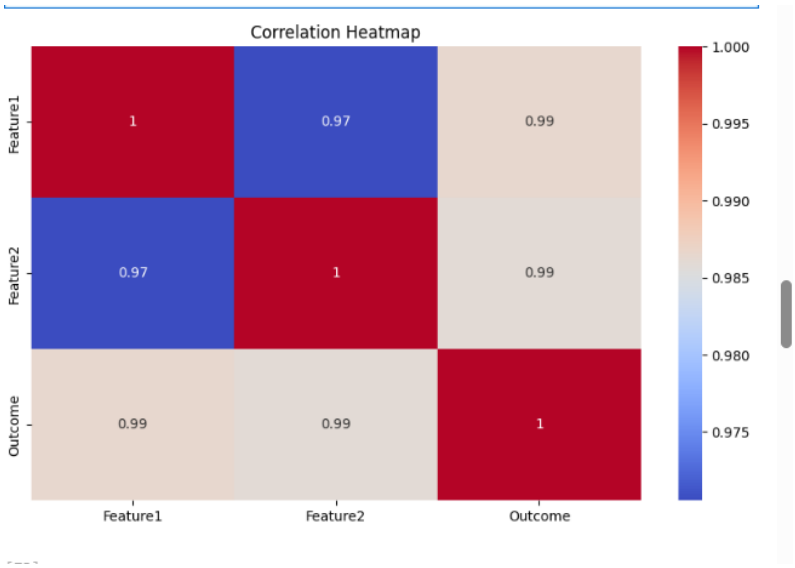
```
plt.figure(figsize=(7,3))
sns.scatterplot(x='Feature1', y='Feature2', data=df)
plt.title("Feature1 vs Feature2")
plt.show()
```



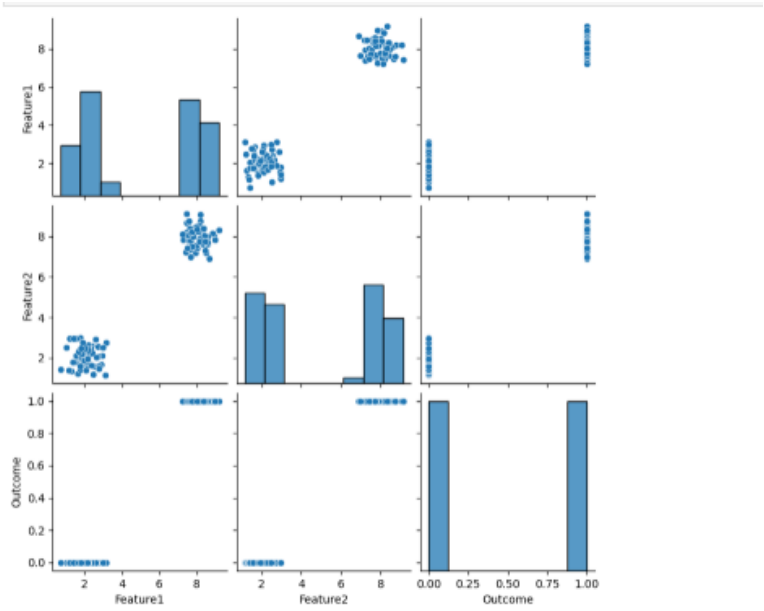

```
plt.figure(figsize=(7,4))
sns.boxplot(x='Feature1', y='Feature2', data=df)
plt.title("Feature1 vs Feature2")
plt.show()
```



```
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```




```
sns.pairplot(df[['Feature1','Feature2','Outcome']])
plt.show()
```

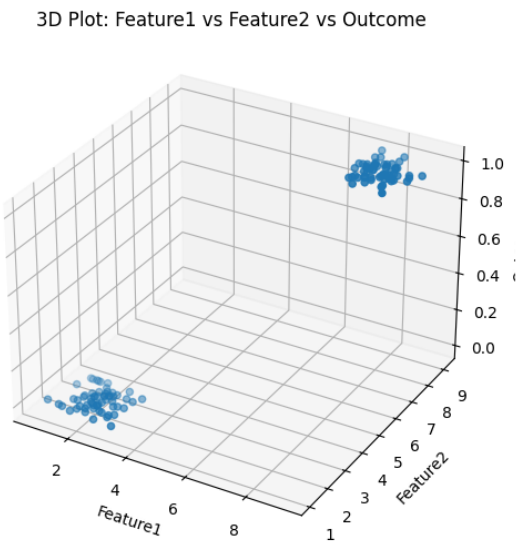


```
print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R²  :", r2_score(y_test, y_pred))
```

MAE : 0.0

MSE : 0.0

RMSE: 0.0

R² : 1.0

```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
df = df.dropna()
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(df['Feature1'], df['Feature2'], df['Outcome'])
ax.set_xlabel("Feature1")
ax.set_ylabel("Feature2")
ax.set_zlabel("Outcome")
ax.set_title("3D Plot: Feature1 vs Feature2 vs Outcome")
plt.show()
```